



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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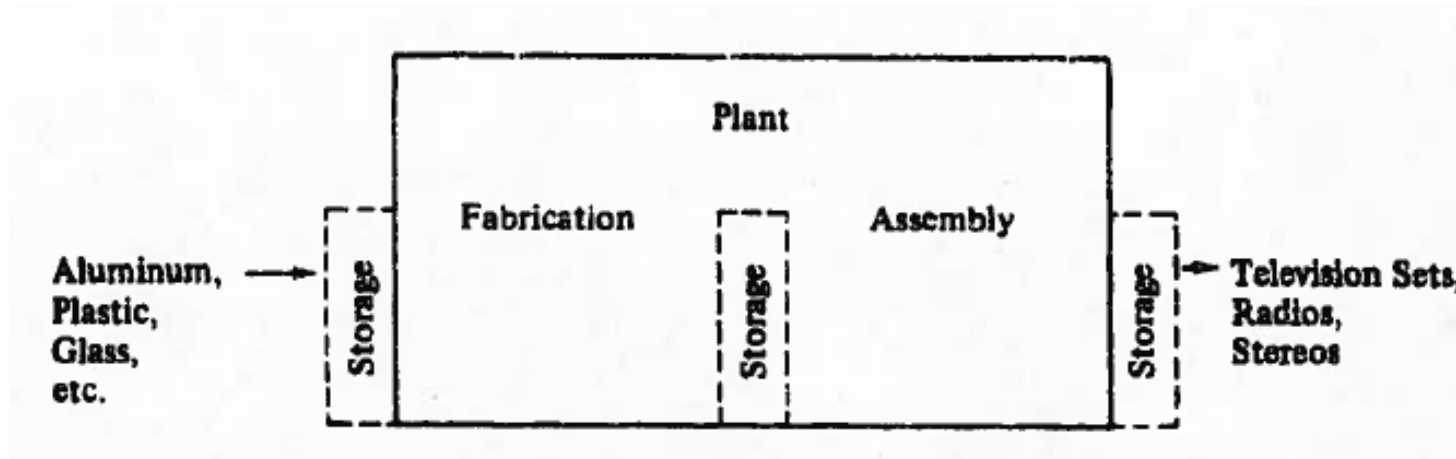


Agenda

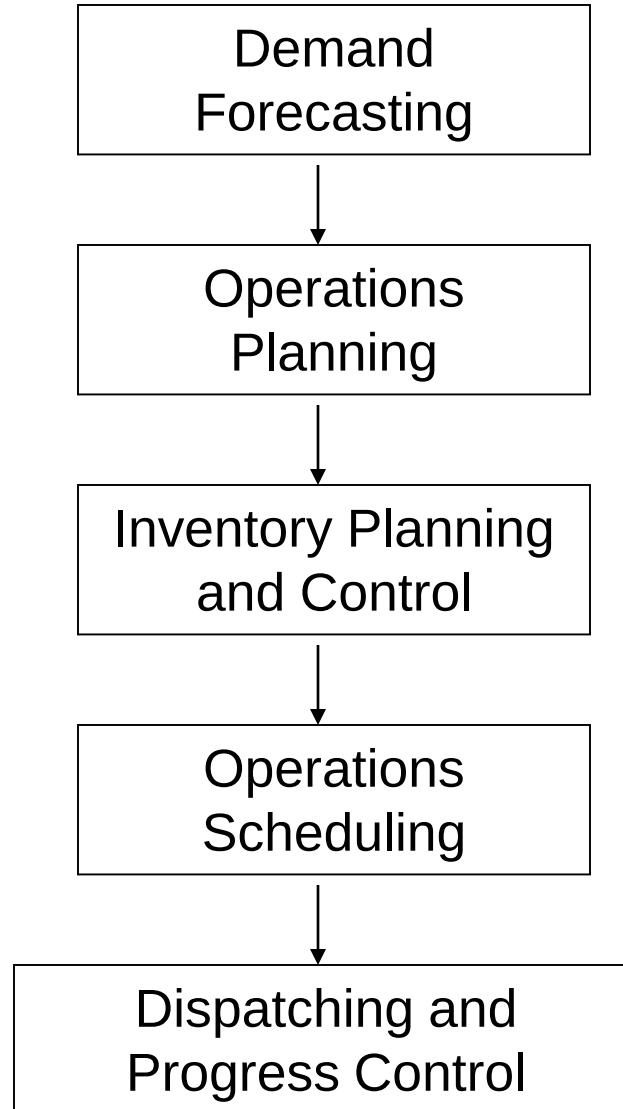
- Chapter 7 Operations Planning and Control

Context

- A plant manufacturing TV sets, radios, etc.
- Purchase materials and parts
- Fabrication of components
- Assembly
- Many configurations of each product
- Several common components in different products
- Common raw materials in different components



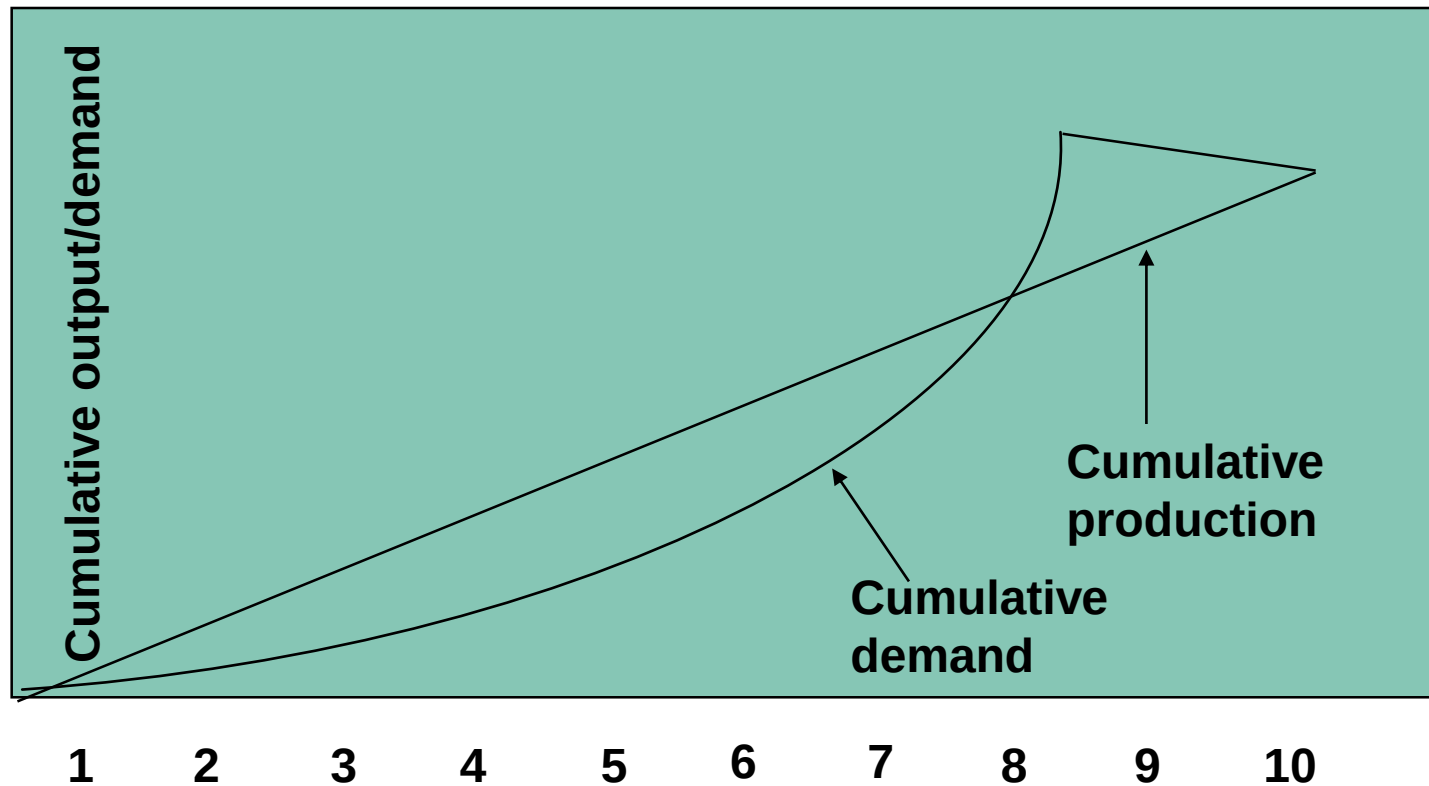
Operations Planning and Control



Operations planning

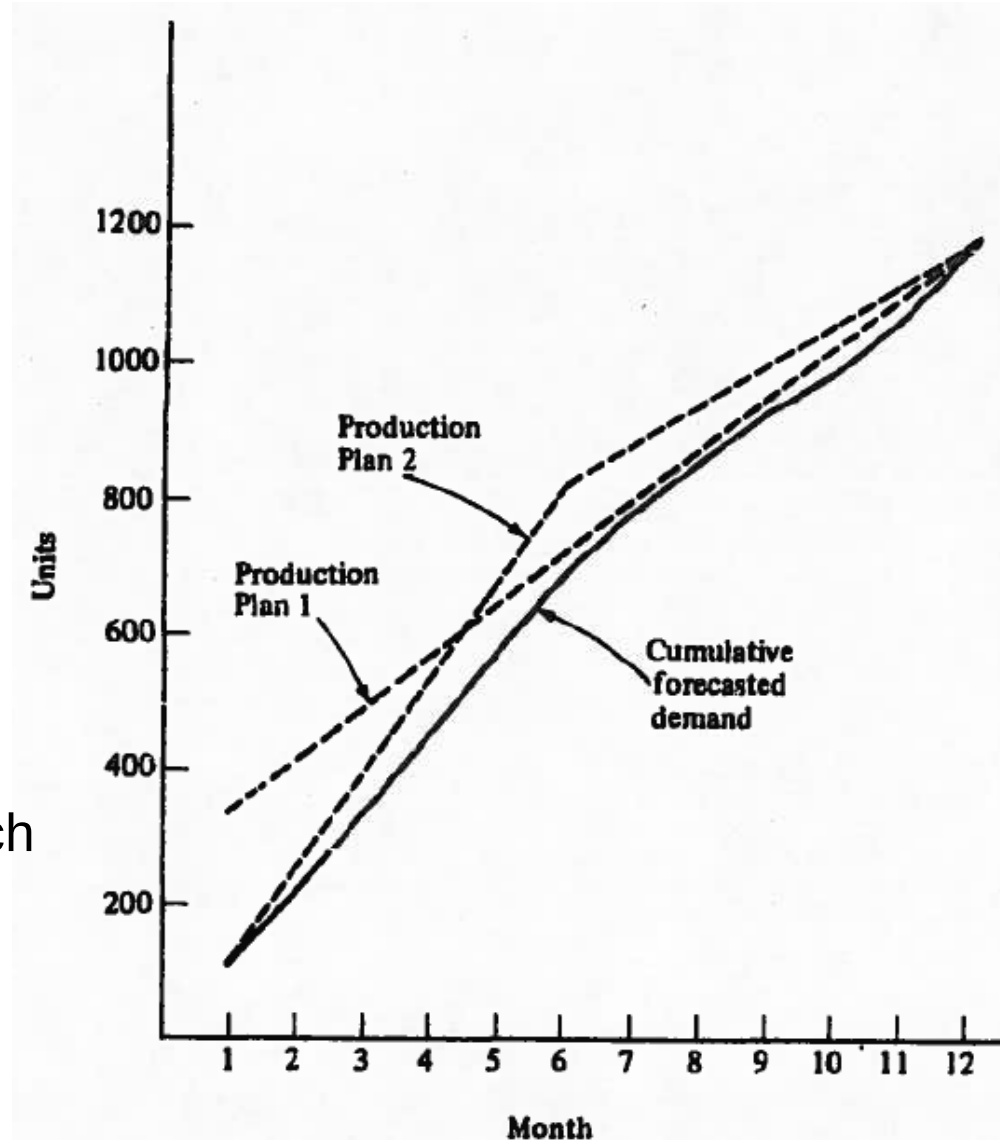
- All resources required are at the right place at the right time and in the needed quantities
- Minimizing the cost
- Budget limits and employment practices imposed by higher management

Cumulative Graph



Planning Strategies

- Maintain a level workforce
 - Over-time (high demand)
 - Idle-time (low demand)
 - Inventory carrying cost
 - Maintain a steady output rate
- Match demand period by period (chase)
 - Frequent hiring/firing
 - High cost of training
- Use a combination of decision variables
 - The cumulative demand in each period must be satisfied.
 - Plan 1 (level) : high initial inventory, no end inventory
 - Plan 2 (chase): less initial inventory, no end inventory



Inventory Planning and Control

- ***How much*** to order (produce)?
 - Order quantity
 - Cost of ordering too much or too little at one time
- ***When (how often)*** to place the orders?
 - Order points
 - Cost of ordering too frequently or not frequently enough

Inventory Planning and Control

- Inventory planning
 - Inputs:
 - Approved operations plan (time-phased outline of plant's activities)
 - Outputs:
 - Net, time-phased requirement for components, parts, raw materials, sub-assemblies, and supplies
- Inventory control
 - Inputs:
 - Inventory plan
 - Sales orders and material requisitions from dispatching
 - Outputs:
 - Order quantities, re-order points, safety stocks
 - Manufacturing batch-sizes

Benefits of Holding Inventory

- Improve customer service
- Reduce costs:
 - ordering costs
 - stockout costs
 - acquisition costs
 - quality related costs
- Efficient and effective operation of production system

Drawbacks of Holding Inventory

- Certain costs increase :
 - carrying costs
 - large-lot quality cost
 - obsolescence
 - production related problems

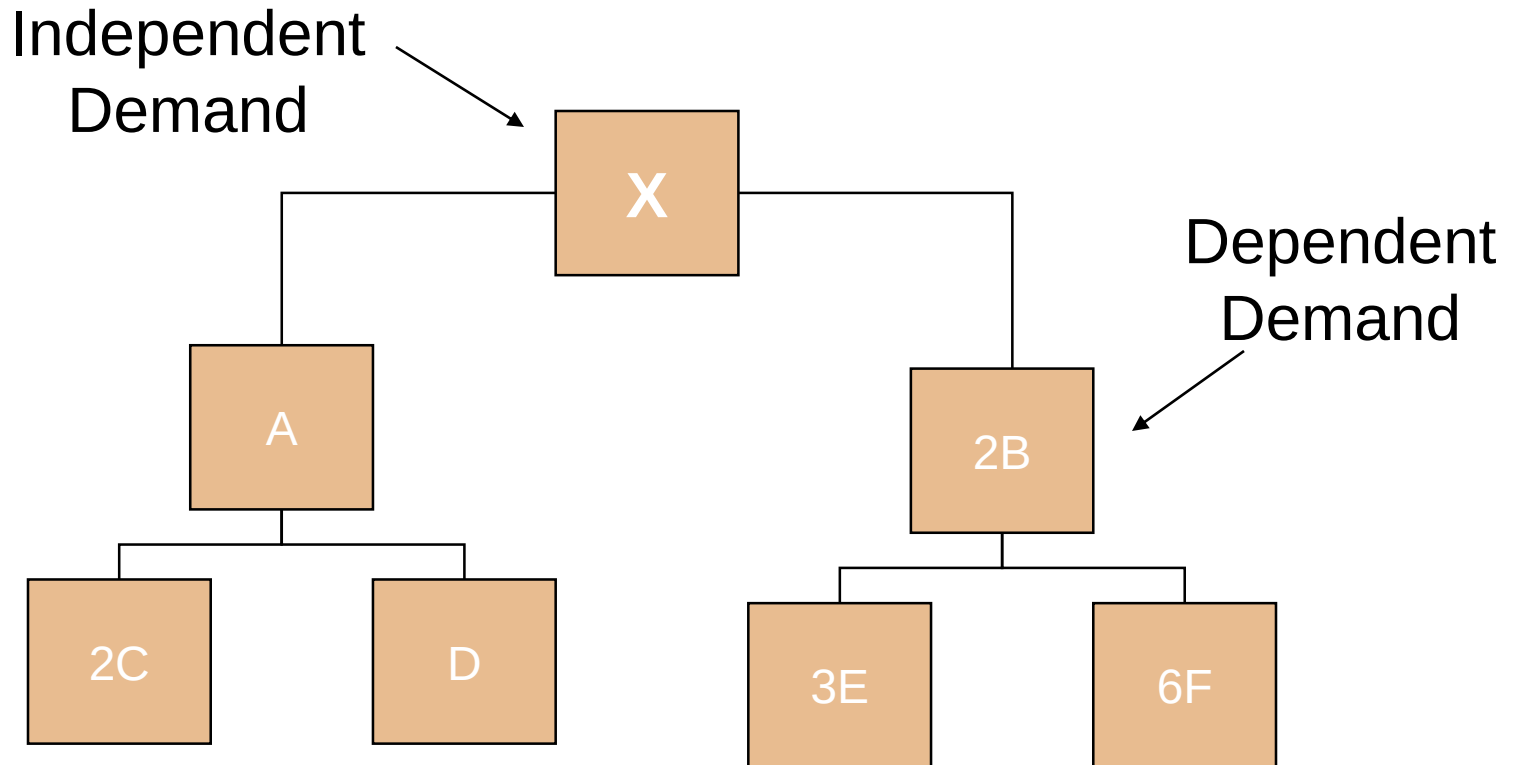
Cost of inventory

- Procurement cost
 - Clerical cost of making-up and processing the order (fixed)
 - Variable cost
- Carrying cost:
 - Cost of money invested in inventory
 - Cost of storage space
 - Cost of obsolescence and spoilage
 - Cost of insurance and taxes

Inventory Systems

- Independent Demand
 - Items whose demand is **independent** of the demand for any other item in inventory (end items)
- Dependent Demand
 - Items whose demand **depends** on the demands for other items (raw materials)

Inventory Systems



Determining Order Quantities

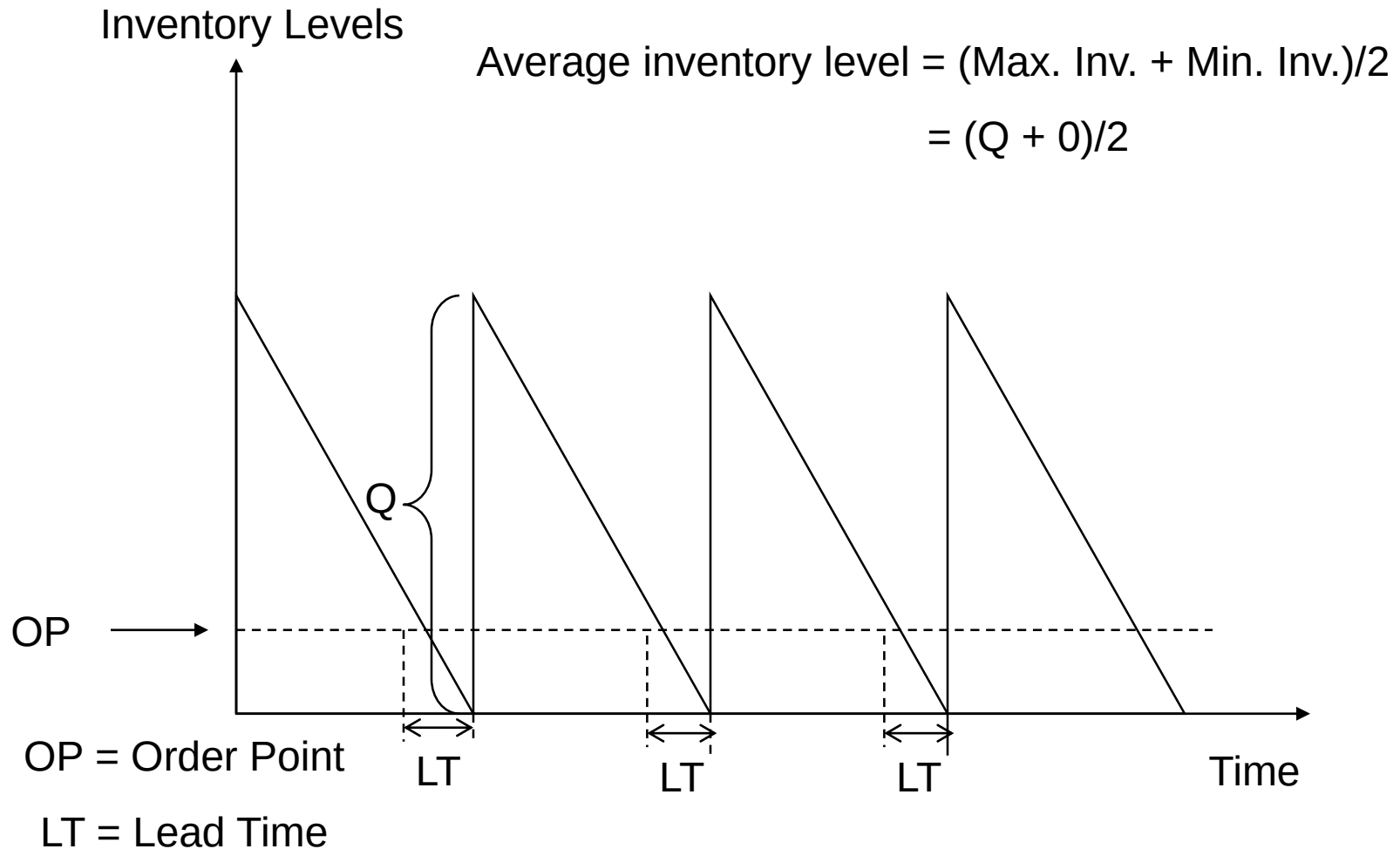
Basic Economic Order Quantity (EOQ) Model

- One type of product
- To decide
 - the order quantity Q of each ordering
 - in every fixed time period (e.g. every week or every 10 days)
 - of a longer time horizon (e.g. a season or a year)
- Required input data or known information:
 - Demand (D) per horizon (year, season),
 - Unit carrying cost (CC) in inventory per horizon (year)
 - Ordering cost (PC) of the batch of products

Assumptions in the classical inventory model

- The usage rate of the item is linear and known with certainty
- The order is received instantaneously when ordered (zero lead time)

Fixed Order Inventory System for Independent Demand



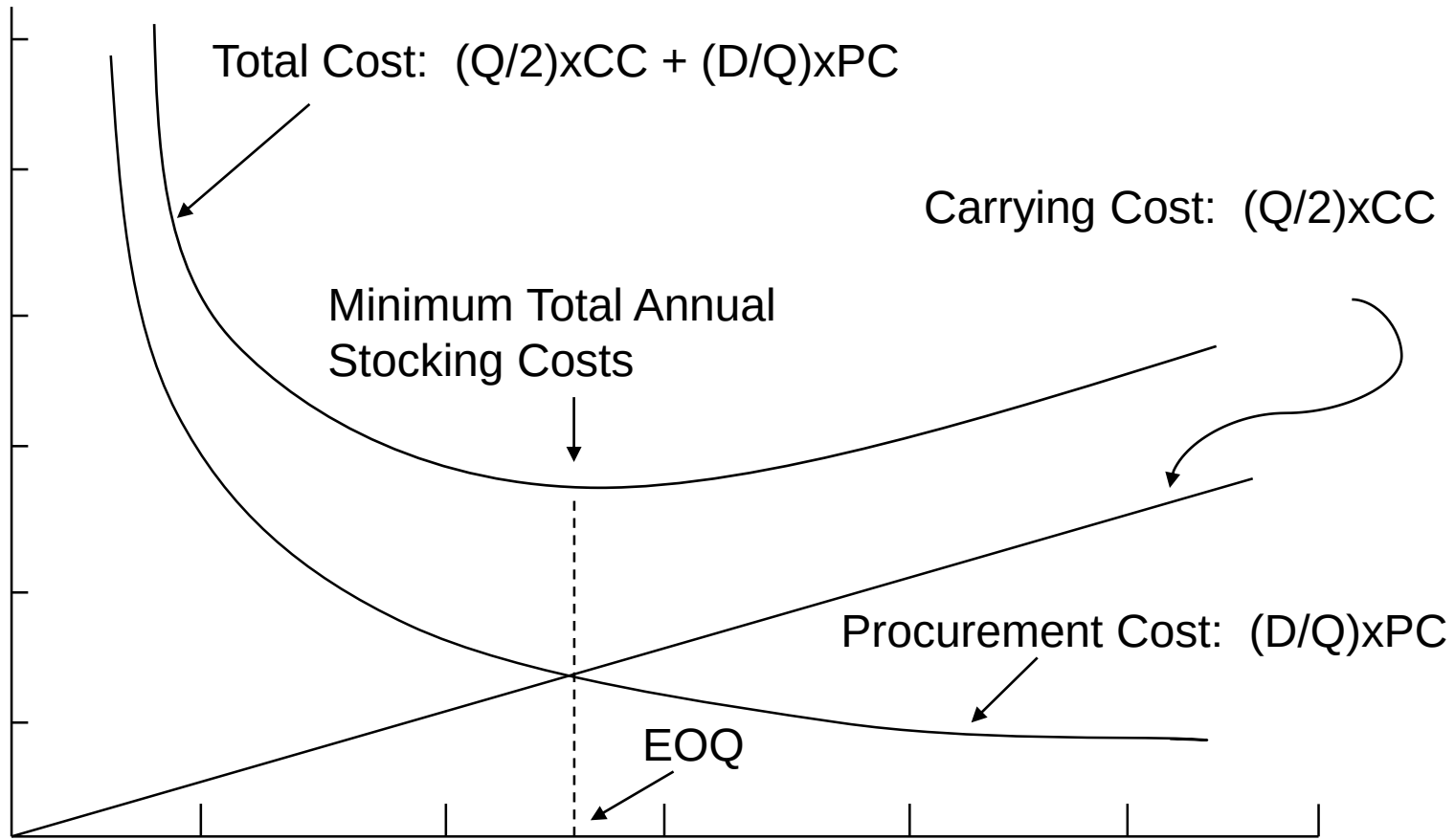
Determining Order Quantities (How Much?)

- Basic Economic Order Quantity (EOQ)
- Assumptions:
 - annual demand (D), carrying cost (CC) and ordering cost (PC) can be estimated
 - average inventory level is $(Q) / 2$ which implies
 - no safety stock
 - orders are received all at once
 - demand occurs at a uniform rate
 - no inventory when an order arrives
 - acquisition cost is fixed, i.e., no quantity discounts
- Annual carrying cost = $(Q/2)CC$
- Annual procurement cost = $(D/Q)PC$
- Total annual stocking cost = $(Q/2)CC + (D/Q)PC$

Inventory Costs

Total cost = procurement cost + carrying cost

Stocking Costs (\$)



Quantity of Material/order (units)

Deriving the EOQ

Using calculus, we take the derivative of the total cost function and set the derivative equal to zero and solve for Q

Total Annual Stocking Costs: $TC = (Q/2) \times CC + (D/Q) \times PC$

$$dTC/dQ = 1/2 \times CC - PC \times (D/Q^2) = 0$$

$$Q^2 = 2 \times D \times PC / CC$$

$$EOQ = \sqrt{\frac{2DPC}{CC}} = \sqrt{\frac{2(\text{Annual Demand})(\text{Order or Setup Cost})}{\text{Annual Holding Cost}}}$$

Example

- Presently, Company ABC uses a rule of thumb to order inventory for widgets. The general manager wonders if the EOQ might save money. The following estimates were calculated from accounting information:

$D = 10,000/\text{year}$

$Q = 400$ per order (present order quantity)

$CC = \$0.40$ per widget per year

$PC = \$5.50$ per order

$$EOQ = \sqrt{\frac{2(5.5)(10000)}{0.4}} = 524.4 \approx 525$$

$$TC = 5.5 \times \frac{10000}{525} + 0.4 \times \frac{525}{2} = 209.76$$

Example

- Presently,
 - ABC orders 400 units each time
 - ABC places 25 ($10,000/400=25$) orders each year.
 - $TC = 5.5 \times 25 + 0.4 \times 400/2 = 217.5$
- EOQ optimal solution suggests
 - ABC order 525 units each time
 - ABC place 19 ($10,000/525 \approx 19$) orders each year
 - $TC = 5.5 \times 19 + 0.4 \times 525/2 = 209.5$

EOQ Model variations

- Issues to consider and the model needs to be revised to reflect
 - Stockout, customer responsiveness, and related costs
 - Quantity discounts and variable acquisition cost
 - Demand variations
 - Graduate inventory built-up
 - Others